

Quantile Regression and GAMLSS

Forty-four chapters have been spent on one number. Least squares estimates a conditional mean; the generalized linear model estimates a transformed conditional mean; splines and additive models let that mean bend. Everything else about the conditional distribution of the response — its spread, its skewness, its tails — entered as an assumption to be checked, a nuisance to be corrected for, or a reason to widen a standard error. This last chapter turns the nuisance into the subject.

There are two ways to do it. One drops the parametric model and estimates the conditional quantiles directly, each level fitted by its own criterion: that is quantile regression, whose loss function is as simple as squared error and as easy to justify. The other keeps a parametric family but lets *every* parameter of it carry its own structured additive predictor: that is GAMLSS. Between them sits expectile regression, which buys a differentiable loss at the price of a less transparent interpretation. The chapter ends where a book about models should end: with how to tell whether one estimated distribution is better than another.

What you need. [Chapter 5](#) and [Chapter 6](#) (the linear model, least squares and the best linear predictor Theorem 6.11.1), [Chapter 21](#) (heteroscedasticity, weighted least squares and the sandwich Definition 21.4.1), [Chapter 22](#) (Definition 22.2.1, the older route to a well-behaved response), [Chapter 23](#) (the bootstrap Definition 23.1.1 and the permutation schemes Definition 23.5.1), [Chapter 25](#) (what a fitted curve does and does not mean), [Chapter 29](#) (cross-validation Theorem 29.3.2, AIC Proposition 29.2.4, selection bias Proposition 29.5.1), [Chapter 30](#) (penalized fitting Definition 30.1.1) and [Chapter 34](#) (the generalized linear model Definition 34.2.1 and its iteratively reweighted least squares Theorem 34.4.1). The last two sections use the penalized splines of [Chapter 43](#) and the structured additive predictors of [Chapter 44](#); the M-estimation machinery of [Section 20.6](#) (Proposition 20.6.1) is the ancestor of two algorithms below. This is a list of where things came from rather than of what must be read first: no proof in the chapter cites an earlier one, and the results named are pointed at in the text where they are recognized again.

45.-1.1 Roadmap

- [Why model more than the mean](#)
- [Quantile regression](#)
- [Inference for quantile regression](#)
- [Expectile regression](#)
- [Models for location, scale and shape](#)
- [Comparing distributional models](#)
- [Summary and notes](#)